

PV Internship

Task 3 Causality Assessment

Evaluate the relationship between a drug and an adverse reaction using standard causality assessment methods. Analyse case studies and classify reactions as Certain, Probable, or Possible.

CASE STUDY

6 yr old male child (named EA) with body weight 17 kg, admitted with history of fever, swollen neck, abdominal pain joint pain and reddened tongue and diagnosed as Kawasaki disease. His lab investigation data:

Hb: 10.0 gm% ,TC: 10900/c.mm Platelets: 2.87 lakhs/c.mm

PCV: 33.5% MCV: 78.8 fl.

Treatment was started on 23rd June 2023 with Inj.Intravenous Immunoglobulin (IVIG) (Mfg by: Reliance life sciences, batch no: ID05G21020, Mfg date: Sep 2022, Expiry date: Aug 2025)

Dose - 2g/kg/day IV OD.

The patient developed itching all over the body and rashes on 23rd June 2023 following the injection. Drug was withdrawn

24th June 2023 and the reaction was managed with Inj .Chlopheniramine IV and Inj.Dexamethasone IV .

He recovered from the reaction on 24th June 2023.

The suspected drug was again given to the patient on 27th June 2023 and itching all over the body recurred the same day.

The drug was stopped and appropriate treatment given.

Suspected drug: Intravenous Immunoglobulin (IVIG)

Adverse reaction Itching and rashes

Date of onset: 23 June 2023

Date of recovery: 24 June 2023

Action taken: IVIG withdrawn

Outcome: Recovered

Rechallenge: Positive — itching recurred after IVIG was re-administered on 27 June 2023.

CAUSALITY ASSESSMENT -Using NARANJO Scale

No.	Naranjo question	Answer	Score
1	Are there previous conclusive reports on this reaction?	Do not know	0
2	Did the adverse event appear after the suspected drug was administered?	Yes	+2

No.	Naranjo question	Answer	Score
3	Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	Yes	+1
4	Did the adverse event reappear when the drug was re-administered?	Yes	+2
5	Are there alternative causes other than the drug that could on their own have caused the reaction?	Do not know	0
6	Did the reaction reappear when a placebo was given?	Do not know	0
7	Was the drug detected in blood or other body fluid in concentrations known to be toxic?	Do not know	0
8	Was the reaction more severe when the dose was increased or less severe when the dose was decreased?	Do not know	0
9	Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	Do not know	0
10	Was the adverse event confirmed by any objective evidence?	Do not know	0
Total Score	5		

Naranjo Score Interpretation

Total Naranjo score = 5

Score	Causality
≥ 9	Definite
5–8	Probable
1–4	Possible
≤ 0	Doubtful

CAUSALITY classification- PROBABLE relationship between intravenous Immunoglobulin and reported rashes and itching.

